

Explainable AI for predictive maintenance: A review and standardized evaluation framework

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ABSTRACT

This research paper investigates the integration of Explainable Artificial Intelligence (XAI) into Predictive Maintenance (PdM) systems, aiming to enhance transparency, interpretability, and reliability in industrial applications. The primary contribution is the introduction of the Explainability Parameters (XPA) framework, which offers a structured methodology for evaluating and applying XAI in PdM. The study systematically reviews recent advancements and challenges in the literature, categorising explanations into pre-modelling, in-modelling, and post-modelling processes. It presents and analyses significant case studies across various industrial sectors to illustrate the practical implications and hurdles of XAI methodologies. Key findings indicate that while XAI significantly improves the effectiveness and trustworthiness of PdM by clarifying model predictions, its implementation is hindered by the complexity of industrial data and the absence of standardised evaluation methods. The XPA framework addresses these challenges by providing tailored metrics for specific applications and advocating for a multi-phase approach to convert technical outputs into actionable maintenance recommendations. The originality of this paper lies in its comprehensive review and the establishment of rigorous standards for assessing XAI methodologies, thereby bridging the gap between theoretical frameworks and practical applications. By promoting adaptable XAI frameworks that cater to real-world industrial needs, this study fosters trust in automated decision-making processes. It enhances the overall understanding of XAI's role in PdM.

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1. Introduction

Artificial Intelligence (AI) algorithms learn from data to predict events or data points that align with users' goals. However, managers and users of these automated systems need to understand the reasoning mechanisms that underlie these predictions, which in turn inform decisions (Vasconcelos et al., 2024). Decisions can lead to major failures or errors in some critical domains, such as healthcare, defence, and finance (Hassija et al., 2024). To avoid these misleading situations, humans must comprehend the AI algorithms' predictions to reinforce trust. Some of the most powerful AI algorithms are black-box algorithms, primarily those based on neural networks (Aghababaeian et al., 2023). A solution might be rule-based systems that served as the basis for symbolic AI algorithms, which used logic to express and infer facts and in which reasoning was explicitly and formally expressed (Zeng et al., 2023). However, this means it is effort- and time-consuming, as it requires coding every case, leading to undecidable problems (Bothra et al., 2024). Another recent and trending approach is XAI (eXplainable AI), which, as its name suggests, aims to explain the rules that underlie data predictions. XAI bridges the gap between human decision-makers, plans, and black-box techniques (Bobek et al., 2023).

Fig. 1 compares the two approaches to artificial intelligence: The top Path is symbolic AI Reasoning, and the Bottom is learning-based AI. Top Path involves data and rules, generating predictions, and making decisions. Bottom Path involves training models using machine learning or deep learning techniques, making predictions on new data, and using XAI techniques to understand how the model arrived at its predictions. This path emphasises trust, suggesting that decisions based on ML/DL models are more trustworthy if we can understand how the model arrived at its predictions. However, decisions based solely on black-box ML/DL models without XAI might not be fully trusted. This figure compares the transparency and explainability of symbolic AI reasoning versus the potential lack thereof in ML/DL models.

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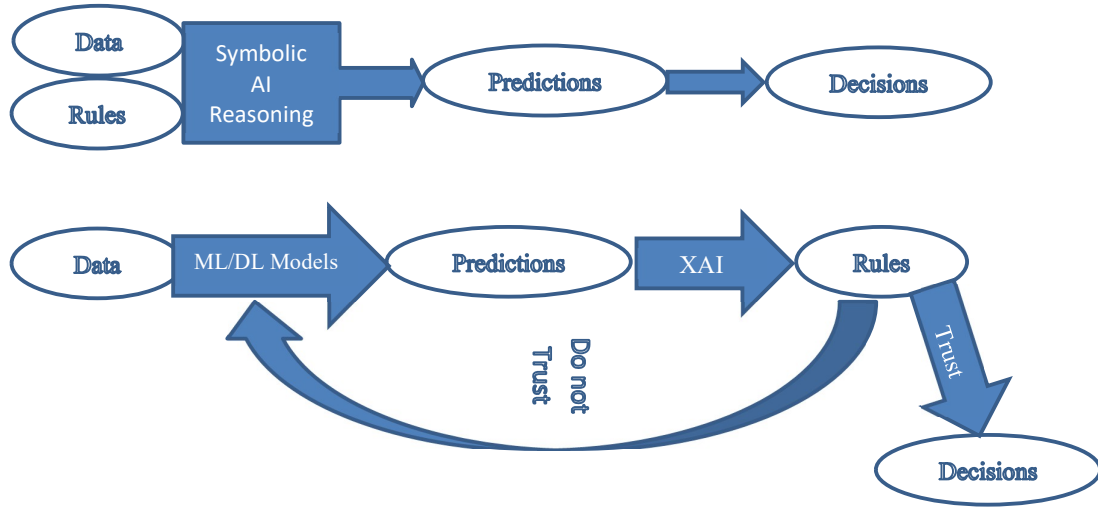


Fig. 1. Symbolic AI versus Learning-based AI

Regardless of the underlying techniques used in XAI methods, their goal is always to explain the model's decisions by describing the relationship between the input and the output. Such a description can be presented in various forms: feature importance, feature impact, decision rules, prototypes, and examples (Kuk et al., 2023).

The term “interpretability” is commonly used in the machine learning literature in contrast to “explainability”. Miller (2019) states that interpretability is the degree to which a human can understand the causes of a decision. Bukart and Huber (2021) argue that interpretability alone is insufficient to address the challenges of comprehending black-box models. It is imperative to prioritise explainability over mere interpretability to establish user confidence and obtain valuable insights into the underlying causes, rationales, and decision-making processes of black-box methodologies. While explainable models are typically interpretable by default, it is essential to note that this is not always the case. An explainable AI system must provide proof, support, or justification for every decision (Zemmouchi-Ghomari, 2023).

The practical application of XAI in industry is hindered by several issues (Rajaoarisoa et al., 2025). A significant obstacle lies in the intricate nature of industrial operations and the opaque data implicated. Industrial data often requires proper labelling, exhibits noise, and needs to be completed; as a result, it lacks explicit contextual information about the underlying processes that generated it. The lack of transparency in AI models makes it difficult to interpret and trust their decision-making processes. In addition, XAI approaches must prioritise knowledge production over mere model debugging. Deriving significant insights from AI models and data is essential for enhancing industrial processes. However, existing XAI methodologies often prioritise elucidating model predictions over producing practical insights (Rane et al., 2024).

Another obstacle arises when attempting to incorporate subject expertise into XAI explanations. In sectors characterised by robust theoretical underpinnings in mathematics or physics, such as the steel industry, integrating domain expertise into AI systems is crucial to ensure high-quality decision support. Nevertheless, current XAI techniques fail to effectively leverage domain expertise, resulting in a lack of understanding of the alignment between AI models and established theoretical principles. Moreover, the reciprocal exchange of information between humans and AI systems presents a difficulty for XAI in the industrial sector. To cultivate reliable explanations and make well-informed judgments based on AI insights, it is imperative to establish effective communication and collaboration between domain experts and AI systems. In industry, the issues associated with XAI primarily pertain to the imperatives of transparency, knowledge development, integration of domain expertise, and facilitating successful human-AI interaction (Chamola et al., 2023). The resolution of these difficulties is crucial in fully harnessing the capabilities of XAI to enhance comprehension and decision-making within industrial operations.

Industries often use black-box AI models to predict equipment failures accurately. However, in a predictive maintenance (PdM) context, managers and operators need to understand the causes of predicted failures to develop solutions that prevent these pessimistic predictions (see Table 1). The reasons are often related to several system components managed separately by potential AI programs. For example, repairing a machine part requires coordinating inventory, logistics, and staff management. In addition, prognostics cannot be separated from diagnostics, which complexifies the context of PdM. In most cases, the context of these predictions needs to be fully formalised; hence, automated reasoning is not applicable.

In PdM, the AI algorithm triggers alerts. If system operators fully understand these alerts, they can make the right decisions. Providing helpful information can improve decision-making in PdM; some examples are presented in Table 1.

Table 1
PdM framework

Dimension	Useful information for PdM	Signification
What Who Where When How	Identifying the component of the system and the phase(s) of the process where the problem occurred	Responsibilities
Why	Understanding the reasons why the problem occurred for design improvement	Root causes
So what	Understanding the severity and the consequences of the equipment failures Selecting an optimal plan for repair from several alternatives	Implications

Despite the numerous works in this field (Arsenault et al., 2025; Infant et al., 2025; Rojas et al., 2025; Mahale et al., 2025; Ayeni, 2025), further research is needed to address the limitations acknowledged by the authors. Most studies on XAI for PdM do not assess whether the data used to train the models can support desired insights. Outcome and end-user explainability are also limited, with only a few studies focusing on understanding why an AI model makes a specific prediction or making AI explanations understandable to non-technical users (Kisten et al., 2024; Haque, 2025).

Research should combine global explainability with model-specific and model-agnostic methods to provide more comprehensive explanations (Belle & Papantonis, 2020; Wang, 2024). Developing explanation metrics — such as safety, security, cost, and time — is crucial to ensuring relevant, practical explanations for PdM applications.

One of the primary gaps is the absence of robust explanation assessment measures in existing literature, which often overlooks the need for a coherent framework to evaluate the effectiveness of XAI explanations in PdM applications. Additionally, current methodologies tend to focus on debugging model predictions rather than generating practical insights for domain experts. The paper also highlights the limited scope of existing research, which primarily emphasises model parameters without adequately addressing outcome and end-user explainability. Furthermore, it identifies a critical need for improved uncertainty management in XAI for PdM, which is essential for fostering user confidence in the face of unforeseen or adversarial scenarios.

The paper's contributions are multifaceted and significant. Firstly, it introduces the Explainability Parameters (XPA) framework, which provides a novel set of rigorous metrics for evaluating XAI implementations in PdM. This framework includes parameters such as Depth of Explanation, Predictive Accuracy, and Stability, tailored to different maintenance application types, ensuring meaningful evaluations across various industrial contexts. Secondly, the paper proposes a comprehensive methodological approach that outlines a structured implementation blueprint for utilising XAI in PdM. This approach encompasses data preprocessing, model training, and failure diagnosis, emphasising the critical step of translating technical outputs into human-interpretable, actionable maintenance recommendations. Lastly, the paper offers a holistic analysis of trends and challenges in PdM-XAI, bridging theory and practice by proposing adaptable frameworks that deliver practical insights.

The paper is structured to enhance clarity and focus, particularly by introducing a dedicated Section 6 that consolidates the novel performance metrics and methodological framework. This section leverages insights from earlier sections, transforming critiques into a prescriptive and evaluative framework that advances the field of XAI in PdM. By synthesising findings from a systematic review and case studies, the paper not only critiques existing literature but also provides a tangible proposal for improving the integration of XAI in predictive maintenance, ultimately aiming to enhance transparency, trust, and actionable decision-making in industrial applications.

2. Predictive Maintenance (PdM)

Before delving into PdM, let us examine some terminology, particularly the definitions of anomaly, fault, and failure (Cerquitelli et al., 2021). An anomaly is any significant deviation from the norm. A fault is any unpermitted deviation. A failure is a fault that permanently disrupts the system's function.

There are three types of maintenance from the ISO standards (Hitchcock, 2006), namely:

- Reactive maintenance consists of repairing after a failure occurs.
- Preventive maintenance consists of periodic repairs or replacements based on historical data; in this case, finding a balance between avoiding failures and wasting resources is crucial.
- Predictive maintenance consists of performing maintenance at the exact time determined by analysing health indicators' component or subsystem degradation curves. There are two primary methods for achieving this: model-driven and data-driven.
 - Model-driven methods require skilled labour; some examples of required tasks are capturing the dynamics of moving parts, effects of cracks on measured vibrations, and partial discharge in electrical circuits.

- Data-driven methods automatically use historical data collected from sensors to find relationships between data and outcomes. This method is low-cost but without a performance guarantee, and it is applied in uncertain areas such as transportation systems.
- In some cases, hybrid approaches are used. Some AI systems use experts' knowledge and collected data.

In the rest of this study, we will focus on data-driven models.

Maintenance is divided into two categories based on the time of interest: diagnostics, which focuses on what is happening now, and prognostics, which looks into the future.

2.1. Diagnostics

Diagnostics consists of anomaly detection, fault detection, fault isolation, and fault identification. Each task builds upon the preceding one. Anomaly detection identifies any operation deviation from the norm (Kaur et al., 2018). The next step is to verify whether it is a fault. Then, the fault is isolated (the responsible component), and finally, the fault's severity is estimated, for example, by measuring the crack magnitude.

Anomaly detection is performed by comparing data from networked sensors with historical data or with data from similar equipment. If deviations are identified as expected, an alert is triggered indicating anomalies in the system's functioning.

Unsupervised machine learning algorithms are often used to detect anomalies and are not explainable. Explaining the root causes of detected anomalies remains an open research issue.

Detected anomalies can be recognised as symptoms of a faulty state or of unusual behaviour. Early fault detection prevents problem escalation by enabling maintenance to be performed early while the system is still in an acceptable state. To be effective, fault detection should occur quickly to enable prompt maintenance actions. It should also identify fault types.

In fault isolation, the nature and location of the fault are expected to be identified, which can be challenging. Several faults can have the same symptoms but different underlying root causes. The causes can include degradation over time, accidents, misuse, or operation under unfavourable external conditions. Identifying fault causes often relies on the expertise of the application domain. This can be resource-intensive and time-consuming.

Fault identification involves precisely specifying the fault's severity and size. It aims to quantify the impact of the identified fault on its environment. Other metrics can also be helpful, such as the magnitude and the frequency of occurrence. These data are crucial for informed operational maintenance decision-making, such as determining whether to repair or replace affected components.

Prediction models can fail to detect fault severity because these faults are rare and the model has been trained on a limited number of rare failure causes. The two classes of faulty and healthy cases need to be balanced. This necessitates the development of hybrid methods that integrate data-driven and model-based approaches to inform decision-making.

2.2. Prognostics

While diagnostics concerns the equipment's current health, prognostics concerns its future condition. It forecasts how the equipment degrades over its lifetime and when the degradation is expected to occur (Vogl et al., 2014; Djeddi, 2024).

Prognostics include RUL (Residual Useful Life) estimation, survival analysis, and prescriptive maintenance. The RUL of a system is defined as the interval from the particular time of operation until the end of the system's useful life when it is unable to function. It is based on a mapping between sensor observations and the health condition, then predicting future deterioration of that health; in other words, it is based on the current state of the system and its degradation trends. There are three approaches for RUL estimation:

- A. Mapping between a set of sensor inputs and RUL
- B. Mapping between the approximated health index of degradation level and RUL
- C. Similarity-based matching with or without training a mapping function

Routine preventive maintenance can lead to high costs when replacing regular system components; on the other hand, replacing these components when failures occur can result in even higher costs, including downtime and repair. The RUL estimation enables replacing components before they are likely to fail, thereby optimising resource allocation and ensuring optimal performance. However, RUL estimation is challenging since it relies on high-quality historical data on system operations, maintenance, and failures. Also, equipment degradation is influenced by many factors, such as environmental conditions, leading to variability in the RUL of identical systems under similar operating conditions. It is important to note that RUL estimation for the current industrial asset is based on controlled experiments conducted under simulated operational conditions. The simulation time data and the unseen test data are from the same population. This may only hold for simple machines operating under varying conditions. Indeed, validating RUL predictions requires waiting until the end of a system's life to compare the actual data with the predicted RUL. Survival analysis calculates the probability distribution of equipment survival. It calculates the probability of equipment breakdown over time and estimates the rate of equipment

deterioration during its lifetime. These methods consider a population-level perspective. They do not look at an individual system in operation. They consider the bigger group and the individual system part of this group. This allows us to abstract away from specifics and focus on high-level concepts.

In prescriptive maintenance, AI algorithms make predictions, control the system, and plan to address faults and failures. Depending on the application, these predictive maintenance systems handle personnel scheduling, efficient replacement ordering, and cost management for materials and emissions. Indeed, prescriptive maintenance is an algorithm that optimises total costs by considering multiple factors. The algorithm's output is a course of action that includes resource management, decision-making, and recommendation systems. Thus, incorporating experts' knowledge is essential, and purely data-driven models still need to be more feasible.

3. Explainable Artificial Intelligence (XAI)

Section 3 introduces XAI, a crucial domain that explains data predictions and bridges the gap between human decision-makers and complex AI techniques. It aims to establish a rigorous conceptual framework for XAI, defining core properties like interpretability, transparency, dependability, trust, and intelligibility. XAI methods describe the relationship between input and output, explaining model decisions. The section details XAI's application domains and presents a taxonomy of approaches. It also discusses the obstacles and challenges impeding its full acceptance in industrial settings.

3.1. Definitions

Adadi and Berrada (2018) Examined the significance of XAI across various contexts, highlighting the heterogeneity of explanation objectives within AI systems. They introduced the frameworks of "explain to justify", which focuses on providing justifications for AI decisions to uphold fairness and ethics, and "explain to control", which emphasises enabling users to understand and manage AI systems. Additionally, they discussed "explain to discover", which facilitates learning and understanding of AI decision-making processes. Core properties of XAI are defined to enhance transparency, reliability, and trust in complex models, particularly in PdM (see Fig. 2). The distinct definitions include (Pashami et al., 2023; Arsenault et al., 2025):

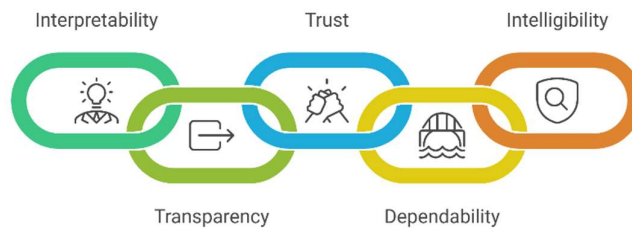


Fig. 2. Foundations of eXplainable AI

- **Interpretability:** The degree to which a human can understand the cause of a decision, operationally defined as the ability to explain in understandable terms, predict model results consistently, and the inherent transparency of a model's operations.
- **Transparency:** Refers to how AI methods obtain their output, essential for responsible AI frameworks, minimising malfunctions, and enabling explainable decisions.
- **Trust:** The confidence users have in AI outcomes, crucial for practical usage. Trust is built when users understand AI recommendations, and a lack of explainability can hinder adoption, especially in regulated industries.
- **Dependability (Reliability and Safety):** The requirement for AI systems to function as intended and safely, enhanced by XAI through justifying decisions and detecting issues.
- **Intelligibility:** The quality of an AI system being clear and understandable, necessary for ensuring transparency and trust, and addressing the black-box problem.

3.2. Application domains

In their study, Adadi and Berrada (2018) examine the multifaceted sectors in which XAI might yield substantial advantages. They also emphasise the variations in the implementation of XAI across different fields. The importance of XAI depends on the specific field and the potential ramifications of AI mistakes. For instance, in fields such as targeted advertising, where mistakes have minimal impact, a relatively low level of interpretability may be sufficient. Conversely, greater comprehensibility is crucial in healthcare diagnosis systems, where errors can have a profound impact on patients (Awotunde et al., 2023). Additionally, the section proposes that domains characterised by strong resistance to errors offer promising opportunities for implementing XAI methodologies. Furthermore, the article highlights government and image identification, where XAI can effectively address significant obstacles and enhance decision-making effectiveness. The distinctions mentioned above underscore the importance of tailoring XAI solutions to meet the specific needs and potential risks of various application domains. This guarantees transparency, trustworthiness, and efficacy in AI systems.

3.3. Types of Explanations

Several authors have proposed taxonomies to categorise XAI techniques, providing a framework for understanding these techniques.

Arrieta et al. (2019) distinguish between explainable models (interpretable ML) that are designed to be explainable and those that are made explainable through post-hoc methods (explainable ML). Gilpin et al. (2018) propose an alternative taxonomy based on three concepts: how DL processes data, how data is represented internally, and how the model was designed to simplify interpretation.

The taxonomy of XAI approaches, as described by Adadi and Berrada, encompasses several techniques, including visualisation methods such as heatmaps, feature significance plots, and decision trees, as well as information extraction methods. Specific methodologies prioritising extracting interpretable rules or patterns from opaque artificial intelligence (AI) models include rule extraction, decision rule learning, and symbolic rule generation. Impact: To assess the relative significance of each feature in the decision-making process, several methods are used to examine the influence of input features on the model's predictions. These methods include sensitivity analysis, feature attribution, and gradient-based approaches. This category encompasses explanations that offer examples or instances to illustrate how the AI model arrived at a given prediction. Examples of such explanations are counterfactual explanations (Wachter et al., 2017), prototype-based explanations, and instance-based reasoning.

According to Angelov et al. (2021), the ontology and taxonomy of XAI (see Fig. 4) at a high level can be detailed below:

- **Transparent model:** Typical transparent models include kNN, decision trees, rule-based learning, Bayesian networks, etc. The decisions made by these models are often transparent, although transparency is not a sufficient guarantee that a model will be readily explainable.
- **Opaque model:** Typical opaque models include random forests, neural networks, and SVMs. Although these models often achieve high accuracy, they are not transparent (see Fig. 3).
- **Model agnostic:** Model-agnostic XAI approaches are designed to be generally applicable. As a result, they must be flexible enough not to depend on the model's intrinsic architecture; they should operate solely by relating a model's inputs to its outputs.
- **Model-specific:** Model-specific XAI approaches often take advantage of knowing a specific model and aim to bring transparency to a particular type of one or several models.

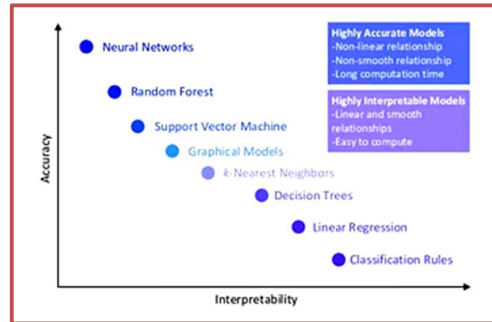


Fig. 3. ML algorithms according to interpretability and accuracy

- **Explanation by simplification:** By simplifying a model via approximation, we can find alternatives to the original models to explain the prediction we are interested in. For example, we can build a linear model or a decision tree around a model's predictions, using the resulting model as a surrogate to explain the more complex one.
- **Explanation by feature relevance:** This idea is similar to simplification. Roughly, this type of XAI approach evaluates a feature based on its average expected marginal contribution to the model's decision, considering all possible feature combinations.
- **Visual explanation:** This type of XAI approach is based on visualisation. As such, the family of data visualisation approaches can be exploited to interpret the prediction or decision over the input data.
- **Local explanation:** Local explanations approximate the model in a narrow area around a specific instance of interest and offer information about how the model operates when encountering inputs similar to the one we are interested in explaining.

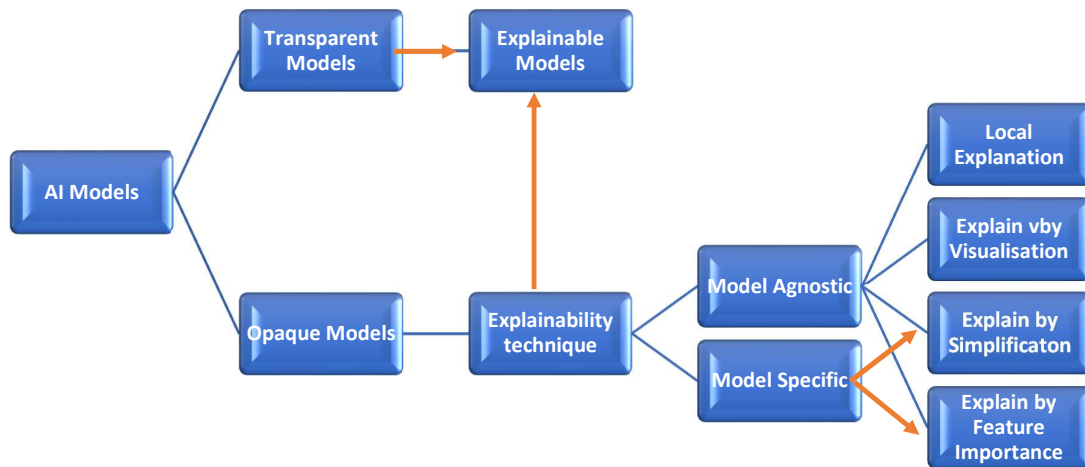


Fig. 4. The high-level ontology of XAI approaches

Generally, XAI techniques can be categorised into pre-modelling, in-modelling, and post-modelling.

3.3.1. Pre-modeling

Pre-modelling techniques focus on understanding data through exploratory data analysis methods, with assistance from domain experts (see Fig. 5), to draw insights and accurately interpret the domain (Geburu et al., 2021). One known technique in this category is data summarisation. It is a technique used in the initial phase of data preprocessing to assess the importance of features based on the goals of ML tasks. Zhang et al. (2021) reported that visual analytics-based approaches help maintenance log analysts extract and explain patterns specific to some issues. Data summarisation, data visualisation, and analytics aim to verify the alignment between existing domain knowledge and knowledge derived from data. Clustering algorithms can be used for conformance checking to derive interpretable causes of machine malfunctions.

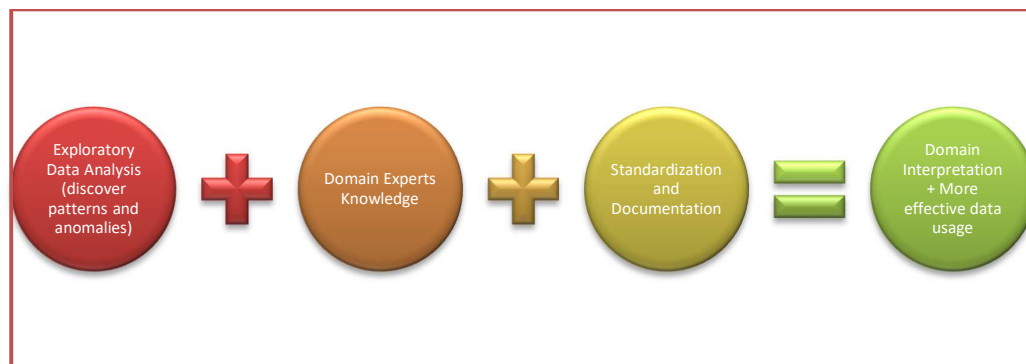


Fig. 5. Data Explainability for Pre-modelling

Other techniques include tabular data description methods (Bobek et al., 2022) and time-series data modalities (Hajgato et al., 2022).

3.3.2. In-modeling

In-modelling models aim to provide explanations by selecting the appropriate algorithm complexity. Most ML algorithms are black-box models and do not enable interpretations. However, some techniques are considered interpretable; the most common example is decision trees; if the tree is small, it is easy to follow the reasoning

pathway from one parameter to another. Unfortunately, simplicity comes at the cost of relatively low predictive performance. Another example is linear regression, in which each parameter is associated with a weight, the cumulative weighted sum being the final prediction. The weights enable each parameter to correspond to its importance in the prediction. However, these algorithms may perform better for nonlinear tasks, as they are more complex (Bach et al., 2015).

a. Physics-inspired models: These models are based on mathematical equations describing a system's behaviour. A trade-off is necessary to balance the simplicity of equations, ease of understanding, and high performance with complex equations (difficult to understand). Additionally, the complexity of a predictive maintenance system makes it challenging to model its overall behaviour. Only some components can be represented using physical formulations. Hence, developing a PdM system purely based on physical equations is unfeasible. A hybrid approach combining physical and data-driven methods can yield better results than a single-based approach. For example, the physics model can serve as a regularizer for the parameters in an ML model's loss function. However, these models are crucial because humans, particularly technicians and managers, must first understand the underlying physics and mathematical equations to understand the predictions.

b. Causal models: "They are designed to make explicit the role and importance of each component of the ML model on its decisions with concepts from the causality" (Moraffah et al., 2020). In other words, causal models examine how certain aspects of the system influence others. Causality is defined as: "the extent to which an explanation of a statement to a human expert achieves a specified level of causal understanding with effectiveness, efficiency, and satisfaction in a specified context of use" (Holzinger et al., 2019). Causal models describe the consequences of various alternatives, which is very important for decision support systems.

3.3.3. Post-modeling

The most developed solutions of the XAI discipline fall into this category. They employ several techniques for feature relevance, including representative data samples and counterfactual examples.

a. Explanation by example: the example-based explanation is based on providing a data instance to explain the prediction; prototyping can help to provide an overview of a dataset based on some representative examples; however, because of their limited number, they might not capture the dataset variations. One of the techniques used for prototyping is k-medoid clustering (Kaufmann & Rousseeuw), which finds prototypes that are the most representative of the clusters in the data. Another method is the MMD-critic (Kim et al., 2016), which finds prototypes and critics by minimising the maximum mean discrepancy between the prototype's distribution and the data distribution. The criticism examples are the instances that maximise the witness function, capturing data points at which the distribution of the prototypes and data diverges.

b. Feature importance: This is the most popular method in XAI research. The idea is to assign each feature a weight that describes its impact on the model output. When the weight is near zero, it means there is little to no impact. In some cases, we can find negative weights. For example, the sign indicates in which direction the feature moves the prediction, decreasing/increasing the RUL of a component.

One of the methods used for this is LIME, which stands for Local Interpretable Model-agnostic Explanations (Ribeiro et al., 2016). It is a model-agnostic approach that can be used with any black-box model. The idea is to perturb the input instance to generate a new dataset, which is used to train and interpret the model (e.g., linear regression to assign feature importance values). Its drawback is that small changes in data can result in significant changes in feature importance. Hence, we can have different explanations for similar observations.

Another popular method is SHAP, which stands for Shapley Additive Explanations (Hart, 1989; Lundberg & Lee, 2017). The idea is to calculate the contribution of each feature in the prediction by building coalitions of features and randomly perturbing the data to see the impact.

c. Sensitivity Analysis: This technique is similar to feature importance in that it calculates the impact of input features on the prediction. It calculates the gradient of each feature. In some cases, the considered inputs are images; in others, they are time series. A saliency map is created to make predictions based on the inputs using convolutional neural networks. Thus, this technique is primarily used in image processing.

d. Counterfactual explanations: This explanation includes finding an example of each input with the smallest distance from the original. The purpose is to answer the question, “What is the minimum required modification to change the prediction of the model?”. This technique can be seen as an optimisation problem in which we want to minimise the distance between the original point x and the corresponding counterfactual x' while ensuring the new prediction is any class different from the class x . The drawback of this technique is that it depends heavily on the used distance metric; provided explanations vary according to it, even if they are about the same observations. Another drawback is that the feasibility of the explanation is not always possible (the proposed point is within the boundaries of the process). Several methods have been proposed to address this issue, including the incorporation of a training dataset into the optimisation process and minimising the distance between existing points and the explanations. Alternatively, a variational autoencoder can be employed to ensure the feasibility of the generated explanations.

By combining XAI techniques, researchers and professionals can develop comprehensive XAI solutions that address the diverse requirements and challenges of explaining complex AI models across various application areas.

3.4. *XAI challenges*

As identified by Adadi and Berrada (2018), the primary obstacles associated with XAI encompass a dearth of formal problem formulation and definitions, inadequate examination of the human role in explainability approaches, the necessity for interdisciplinary collaboration, and a limited exploration of AI system explainability beyond the interpretation of machine learning models.

From an industry standpoint, the drawbacks of explainable machine learning are complex and present substantial obstacles to the acceptance and efficacy of explainability methods (Verma et al., 2021). A significant concern is the need for universally accepted benchmarks to assess the efficacy of explainability techniques. The absence of well-defined benchmarks poses challenges in comparing various methodologies and consistently measuring their performance across diverse applications and datasets. The need for further progress in the subject is hindered by the challenges researchers and practitioners face in evaluating the dependability and applicability of explainable models. Another significant obstacle is the restricted emphasis on practical insights derived from explanations. Although existing explainability techniques can offer valuable insights into the rationale behind a specific prediction, they often fail to provide practical recommendations for enhancing or adjusting the model based on the provided explanations. This gap hinders the feasible application of explainability in actual situations, as stakeholders may require guidance on effectively tackling challenges identified through explanations. Moreover, there is a need for increased stakeholder involvement in creating and implementing explainable systems. Including end-users, regulators, and domain experts in the design process is paramount in healthcare, finance, and criminal justice, where the consequences of machine learning judgments carry significant weight. The absence of active stakeholder involvement poses a potential hazard in creating explainable solutions that fail to align with the requirements and expectations of individuals who depend on them for decision-making purposes. Moreover, the need for incentives and enforcement for implementing explainable machine learning adds further complexity to the situation. The significance of transparency and accountability in automated decision-making is underscored by regulatory frameworks such as GDPR in Europe. The balance between privacy and transparency in AI models can be achieved through privacy-preserving techniques, such as differential privacy, federated learning, synthetic data, granular transparency, a user-centric approach, and adherence to ethical guidelines. These techniques aim to preserve individual privacy rights while ensuring fairness and accountability. Granular transparency provides explanations tailored to users' needs and technical expertise, while user-centric approaches allow users to choose their desired level of transparency. Ethical guidelines and best practices for data privacy and security are also essential. By carefully balancing these concepts, AI systems that are both explainable and respectful of individual rights can be developed, thereby building trust in AI and ensuring its use for the benefit of society.

The absence of explicit standards and enforcement methods may diminish enterprises' incentive to allocate resources toward explainability solutions. Without concrete incentives and legislative backing, incorporating explainability into machine learning systems may continue to be less important to numerous enterprises. The literature indicates that most PdM works are limited in their ability to calculate feature importance for different tasks, thereby hindering understanding of the model's outcome.

To address the challenges of XAI in the industrial sector, Bobek et al. (2023) propose several strategies. Firstly, they suggest developing context-aware explanations tailored to specific industrial processes, ensuring they are

relevant, interpretable, and actionable. Secondly, a human-centric design approach is recommended, focusing on creating comprehensible and reliable explanations for domain experts and incorporating their feedback to enhance usability. Additionally, the emphasis of XAI techniques should shift towards knowledge generation tasks, aiming to extract valuable insights from AI models to improve industrial operations. Integrating domain information into XAI systems is also crucial, as it allows AI models to be grounded in established theoretical frameworks, thereby enhancing decision support. Furthermore, enhancing human-AI interaction is crucial, necessitating advancements in communication channels and interfaces to facilitate more effective collaboration between domain experts and AI systems. Lastly, developing XAI algorithms that provide practical insights is vital for enhancing performance and decision-making in industrial settings, ensuring that the knowledge derived from AI models is directly applicable to improving processes.

Table 2 summarises the XAI challenges and their respective solutions perspectives, particularly in the industrial sector.

Table 2
XAI Challenges and potential solutions

Challenge	XAI Solution perspective
Problem formulation and definition	Context-Aware Explanations: Develop XAI techniques that provide context-aware explanations tailored to specific industrial processes and domain-specific expertise. These explanations must be relevant, interpretable, and actionable in the industrial setting.
Undefined Human role in explainability approaches Lack of stakeholder involvement and collaboration	Design with a Human-Centric Approach: Develop XAI systems that prioritise human-centric explanations, ensuring their comprehensibility and reliability for domain experts. Human comments and preferences should be incorporated into the generation of explanations to enhance usability and acceptance. Enhanced Human-AI Interaction: Allocate resources to develop techniques that facilitate the exchange of information between people and AI systems in both directions, allowing for seamless communication and collaboration. Enhancing communication channels and interaction interfaces is crucial for improving collaboration and comprehension between domain experts and AI algorithms.
Need for practical recommendations and explanations oriented toward actions.	Knowledge Generation Tasks: Reorient the emphasis of XAI techniques towards tasks that involve generating knowledge, rather than solely focusing on model debugging and verification. The objective is to develop algorithms that can effectively extract valuable insights and practical knowledge from artificial intelligence models, thereby enhancing industrial operations. Practical Insights: Develop XAI algorithms that offer explanations that result in practical insights for enhancing performance, making informed decisions, and improving processes in industrial environments. Ensuring the direct utilisation of knowledge extracted from AI models to improve industrial processes is paramount.
Lack of domain expertise, knowledge integration, and universally accepted benchmarks for assessing the efficacy of explainability techniques	Domain Information Integration: Enhance XAI systems to incorporate domain information into explanations efficiently. It is imperative to ensure that artificial intelligence (AI) models are grounded in established theoretical frameworks within businesses, utilising robust mathematical or physical principles. Leveraging domain expertise can facilitate enhanced decision support.

4. XAI for PdM

To maximise the PdM's usefulness, the research community needs to develop solutions for the types of explanations the PdM needs (Pashami et al., 2023). According to Angelov et al. (2021), the challenges of XAI for PdM are:

- Taxonomy of PdM tasks and the corresponding XAI solutions: the diverse range of PdM tasks, such as anomaly detection, remaining functional life prediction, and root cause analysis, requires tailored XAI solutions. For example, a model predicting remaining useful life might benefit from global explanations that reveal trends and patterns. In contrast, an anomaly detection model might require local explanations to understand specific outlier instances.
- Matching various types of explanations to different purposes and different PdM tasks: different stakeholders, such as domain experts and maintenance technicians, have varying needs and levels of technical expertise. XAI methods must be adaptable to provide accurate and understandable explanations to diverse audiences. For example, a

domain expert may prefer detailed feature importance scores to gain a deeper understanding of the model's decision-making process. At the same time, a technician may benefit from visual explanations, such as heatmaps or decision trees, to quickly identify critical factors.

- The difference between explaining the data versus explaining the model: it is essential to distinguish between explaining the underlying data patterns and explaining the model's internal workings. For example, a model might accurately predict equipment failures. However, if the explanations focus solely on the model's parameters, they may not provide insights into the physical mechanisms causing the failures. Instead, XAI should help uncover the underlying data patterns that drive the model's predictions, such as specific sensor readings or operational conditions that correlate with increased failure risk.

Accordingly, the first need is to identify the fault. AI can detect deviations; however, due to the system's complexity (with many components interacting with one another), identifying the intricate causes requires domain expertise. The second need is to understand the consequences of a fault. This means understanding the severity and extent of the fault, its potential evolution, and the potential implications or damages that may result. This requires considering the broader context of AI systems. The third is supporting operators in creating proper maintenance or repair plans. Indeed, XAI is traditionally used to improve trust in AI systems' predictions. However, XAI requires greater capabilities and can deliver more value to PdM, particularly by understanding the severity, root causes, and implications across a broad range, and by recommending the best actions to prevent faults, as illustrated in Fig. 6.

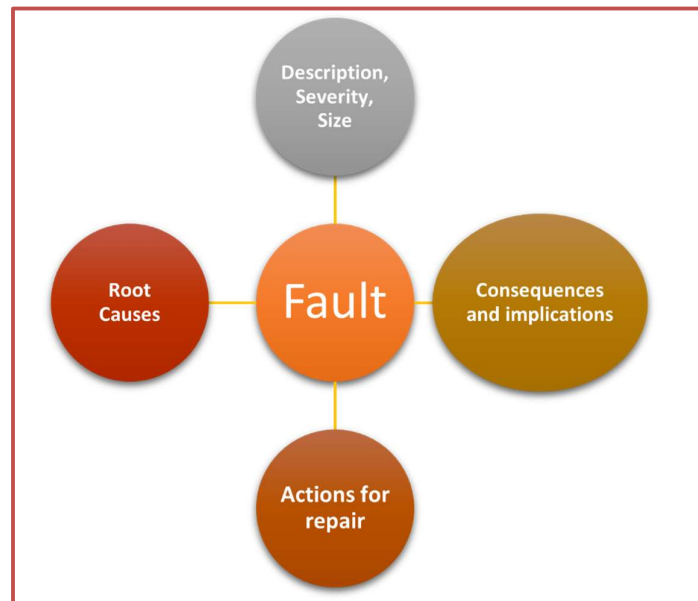


Fig. 6. Fault Context

The typical ML process comprises data gathering, filtering, feature engineering, model selection, hyperparameter tuning, and model evaluation. In the PdM context, most gathered data is sensor data, which is challenging to evaluate manually. Another issue is model selection; the most used models are complex, such as DL, random forests, and support vector machines, which must be interpretable. This means the ML developer needs to understand the model, identify weak points in the ML pipeline, and optimise the solution (Hajgató et al., 2022). This is one of the reasons why model understanding is crucial to improving the reliability and accuracy of the solutions developed. To address this issue, XAI methods might be used to understand the model's behaviour. Predictions in maintenance are not the end of the process. Actions to prevent failures are based on decisions. Under these circumstances, explanations are crucial in helping decision-makers consider the broader context. For PdM actionability (Pashami et al., 2023), three key aspects must be considered: trust, severity analysis, and complex repairs. (1) trust: the expected efficiency benefits can only be achieved if the decision-makers trust the PdM system and its predictions. This is where XAI can play a significant role in helping understand the causes of these predictions and their effects on repair plans. (2) Severity analysis: It is also important to know the impacts of these predictions to prevent collateral damage. XAI can help understand the extent to which the damage or failure will occur. In such situations, decision-makers can gain a broader understanding of the global picture rather than being trapped by silo-based thinking. (3) Complex repairs: In some cases, the repair plan implies the participation of several stakeholders with different priorities and perspectives. Using black-box algorithms for PdM does not facilitate planning for such complex repairs due to a lack of understanding.

However, in addition to the need for XAI solutions, the state of the art must better address these needs. The most popular approaches to XAI are those that provide feature importance but do not address the aforementioned needs. The accuracy of the alerts assesses the effectiveness of PdM systems more than the relevance of the actions that operators or technicians should perform based on the alerts raised by AI systems; XAI solutions should be more focused on actions, as illustrated in Fig. 7.

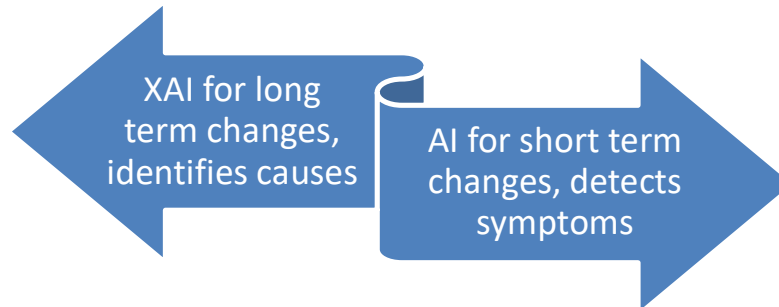


Fig. 7. XAI versus AI systems roles

The case studies section discusses more practical insights and implications on XAI challenges in PdM. Fig. 8 illustrates an overview and summary of XAI for PdM domain concepts.

5. Case Studies

This section presents several case studies that apply XAI techniques to various PdM applications.

- Case Study: Machines in Production** (Hrnjica & Softic, 2020): This case study demonstrates the application of explainable AI in Predictive Maintenance for manufacturing, aiming to avoid unscheduled downtime and associated costs. It showcases the potential of explainable AI through a prediction model and visualisations to support reliable PdM in manufacturing. The experiment utilised a generic dataset, which limited the generalizability of the results. Future work should recommend testing the trained model with real-world manufacturing data from different manufacturers. The used Dataset is a freely available PdM dataset from Azure Blob Storage containing telemetry, errors, failures, and machine properties. The specific XAI technique used is not mentioned in the manuscript. However, the study emphasises the use of visualisations for explainability. The key KPIs for experimentation are Overall accuracy and per-class accuracy of the prediction model. The model achieved an overall accuracy of 99% and an average per-class accuracy of 95%.
- Case Study Nuclear Generation Asset** (Amin et al., 2022): This study investigates using XAI in prognostics and health management (PHM) for a nuclear generation asset, aiming to maximise operational uptime and ensure safety and reliability. It introduces a novel set of algorithms that translate SHAP explanations into text-based, human-interpretable form. This makes the explanations more accessible to stakeholders who may not be experts in machine learning, promoting trust and informed decision-making. The authors note that post-hoc tools, such as SHAP, cannot establish causal links between correlations and physical phenomena. The absence of causality can hinder the reliability and trustworthiness of explanations. The dataset used is not mentioned in the excerpt provided. However, the study indicates using operational data from a nuclear power station. The study demonstrates that SHAP can identify the most important features contributing to the predicted degradation of the nuclear generation asset. The developed algorithms successfully translate SHAP explanations into a more understandable format for non-ML experts.
- Case Study Smart Agriculture** (Kisten et al., 2024): This study addresses the lack of explainability in AI applications in Smart Agricultural Facilities (SAF), aiming to empower farmers to leverage AI's capabilities for PdM fully. It introduces a model that combines XAI with PdM, providing predictive insights and explanations across four dimensions: data, model, outcome, and end-user. This comprehensive approach to explainability promotes transparency, trust, and understanding of AI's decision-making process in SAF. The authors acknowledge that they used only two time series datasets, which limits the scope of the PdM findings. Additionally, the bibliometric analysis was restricted to two academic databases (WoS and Scopus), potentially overlooking relevant literature. Two open-source datasets are used:
 - "Telemetry for Predictive Maintenance": Hourly data intervals from 2015 to 2016, initially with 11 attributes expanded to 40 through feature engineering and refined to 19 for optimal performance.
 - "Pump Sensor": Data recorded every minute from April to August 2018, including data from 52 sensor units before pre-processing. Initially, 54 attributes were distilled to 214, then to 4.

The study leverages SHAP for global and local feature attribution, providing insights into the model's overall behaviour and individual predictions. It also utilises Counterfactual Explanations (CFEs) to explore alternative scenarios and provide actionable insights to end-users. The key KPIs for experimentation are the F1 score, accuracy, and Receiver Operating Characteristic-Area Under the Curve (ROC-AUC). The LSTM classifier demonstrated a 5.81% increase in accuracy, while the XGBoost classifier showed a 7.09% higher F1 score, a 10.66% increase in accuracy, and a 4.29% increase in ROC-AUC compared to previous studies. The study also provided insights into data purity, global and local explanations, and counterfactual scenarios for PdM in SAF.

- Case Study: Automotive Applications** (Krishnamurthy et al., 2020): This study focuses on developing an explainable AI model for predicting wear in automotive applications using predominantly structured images. It presents an explainable AI framework based on a statistical distance metric/pseudometric for wear prediction in structured images. It utilises a feedforward neural network to generate the statistical distance pseudometric, which serves as a predictor in a Bayesian regression model, offering interpretability and validation of the AI model's performance. It does not explicitly mention limitations. However, the study acknowledges that further research is needed to optimise the algorithm and showcase its applications for other defect/damage/anomaly cases (Nor et al., 2022). The used Dataset is not described; however, it mentions using images of wear collected at different periods and wear levels. The study utilises a statistical distance metric/pseudometric generated by a feedforward neural network to explain wear predictions. It draws comparisons with a well-defined Wasserstein-based statistical distance metric to validate the AI model's explainability. The results of the explainable AI model show similarities with the Wasserstein distance-based model, validating the AI model's performance and providing an explainable framework.
- Case Study Aviation Maintenance** (Zeldam, 2018): This case study addresses the challenges of incomplete reporting in aviation maintenance, aiming to improve the completeness and consistency of repair cards through automated failure diagnosis using XAI. It develops an AI-based, data-driven model that automatically diagnoses failures by combining maintenance and usage data. It introduces a new XAI methodology, Failure Diagnosis Explainability (FDE), that provides transparency and interpretability of the assessed diagnosis, enhancing user trust and understanding. The feasibility study was limited by data availability, with only a small amount of failure data being used for training and testing. Additionally, manual variable selection was employed, which may have led to the overlooking of unexpected correlations. The used dataset consists of maintenance and usage data for a central wheel from the RNLAf (Royal Netherlands Air Force) F-16. The maintenance data was used to label failures, and the usage data was incorporated into the diagnostic assessment to account for historical component usage. The XAI Technique used is a Failure Diagnosis Explainability (FDE). FDE explains the assessed diagnosis by comparing the features of a new failure to the expected values of different diagnoses (classes). It utilises a scaled importance factor (χ -factor) to highlight deviations in feature values from their expected values, providing both a visual and a numerical representation of the explanation. The KPI used for experimentation is the accuracy of the failure diagnosis model. The model achieved 81% accuracy in classifying diagnoses, improving by 21% over the baseline. The FDE methodology successfully explained the assessed diagnosis visually and numerically using the χ -factor.

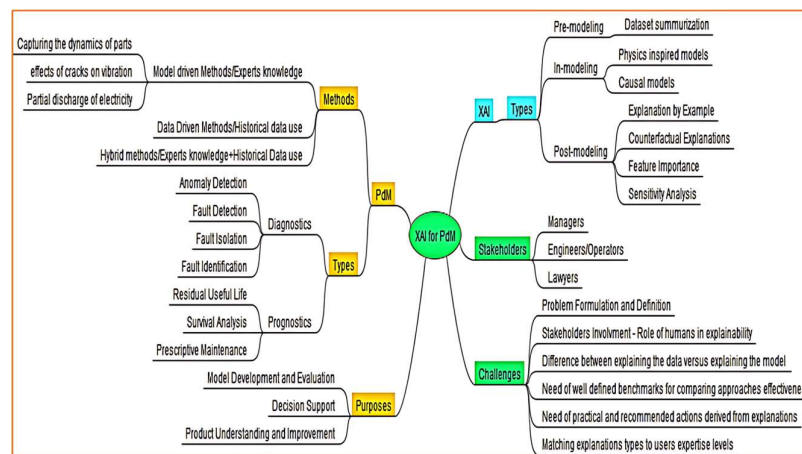


Fig. 8. XAI for PdM mind map

- Case Study Gas Turbine** (Nor et al., 2022): This study investigates the use of XAI in anomaly detection and failure prognostics for a gas turbine engine, aiming to enhance transparency, trust, and understanding of AI models in these critical applications. It combines SHAP and deep learning uncertainty to offer a broader explanation scope, providing insights into the model's decision and confidence level. It also demonstrates SHAP's ability to provide global explanations to enhance the performance of prognostic tasks. The study suffers from a lack of failure data in real-world settings,

as gas turbines are typically well-maintained, limiting the availability of data for model training. The reliance on synthetic data for failure prognostics can introduce biases and limit the generalizability of the findings. Two datasets are used:

- Data from a real-world 18.8 MW twin-shaft industrial gas turbine from Petronas Angsi Oil Platform, recorded over one year (8737 hours), including 98 sensor signals.
- FD001 dataset from NASA Prognostic Centre: Synthetic time series data for 100 turbofan engines, each with 21 sensor signals operating per cycle under three operating conditions, including noise.

The XAI technique used is SHAP, combined with deep learning uncertainty quantification. The experimentation KPIs are anomaly detection rate, prognostic performance metrics (not specified in the excerpt), local accuracy, and consistency of the explanations. The anomaly detection strategy identified 7 of 8 abnormalities, while the best features from the global explanation improved prognostic performance to levels comparable to the best results for the problem. The SHAP explanations were validated through local accuracy and consistency characteristics.

- **Case study Sustainable Urban Water Systems** (Infant et al., 2025): The case study focuses on urban water management challenges, including ageing infrastructure, population growth, and climate change impacts. The aim is to introduce Artificial Intelligence (AI) to improve operational efficiency, resilience, and decision-making by bridging the gap between black-box algorithms and human judgments. The application domain includes hydrological modelling, water demand forecasting, energy optimisation in water distribution networks, and leak detection in water quality monitoring systems. XAI techniques used include SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations), which provide mathematically rigorous global and local interpretations. Counterfactual analysis is employed to explore "what-if" scenarios and optimise system design and resource allocation strategies.

Key performance and value include increased prediction accuracy, energy optimisation, and reduced water loss. However, challenges include the computational intensity of SHAP for large datasets and LIME's limited reliability in highly nonlinear systems. The application also faces socioeconomic obstacles, such as high infrastructure and sensor costs, low digital literacy among local water management workers, and the scarcity of standardised data repositories, particularly in developing cities.

Table 3 compares this case studies based on the available criteria:

Table 3
Case Studies description

Case Study	Application Area	AI Model(s)	XAI Technique(s)	Explainability Focus	Evaluation Metrics	Limitations
Machines in Production	Manufacturing	Not specified	Visualization	Model, Outcome	Accuracy, per-class accuracy	Experimental study, limited sample data
Nuclear Generation Asset	Nuclear power generation	Linear Regression, Random Forest, XGBoost	SHAP, Text-based explanation	Model, Outcome, End-User	R ² , RMSE, MSE	Lack of causality in SHAP explanations; limited intuitive representation of explanations
Smart Agriculture	Smart Agriculture	Convolutional LSTM, BiRNN LSTM, Bagging, AdaBoost	Deepchecks, SHAP, DiCE, Counterfactual Explanations (CFE)	Data, Model, Outcome, End-User	F1 score, Accuracy, ROC-AUC	Limited PdM data; XAI scope limited to data, model, outcome, and end-user explanations; bibliometric analysis limited to two databases
Automotive Applications	Automotive	Feedforward Neural Network	Statistical Distance Metrics (K-S, Wasserstein), Bayesian Regression	Model	Not specified	Limited discussion of the differences between an explainable AI model and a statistical distance metric-based model
Aviation Maintenance	Aviation Maintenance	Random Forest	Failure Diagnosis Explainability (FDE), Box-plots, χ -factor	Model, Outcome, End-User	Accuracy	Limited data; difficulty in data preparation; lack of information on comparing XAI methodologies
Gas Turbine	Gas Turbine Anomaly Detection & Prognostics	Multi-Output Bayesian LSTM	SHAP, CUSUM, Waterfall plots	Model, Outcome	RMSE, Score	Lack of failure data for real-world gas turbine; reliance on synthetic data for turbofan engine prognostics
Sustainable Urban Water Systems	Sustainable Urban Water Systems Engineering (Hydrology, Leak Detection, Demand Forecasting)	ML/DL models (e.g., Neural Networks, Regression)	SHAP, LIME, Counterfactual Analysis, Decision Trees, PDPs, Saliency Maps	Data, Model, Outcome, End-User, Trust, Actionability	Accuracy (e.g., 15% increase), Energy Savings (up to 20%), Water Loss Reduction (up to 12%), RMSE, MAE, R-squared, F1-score	Computational intensity for large datasets (SHAP); reliability limitations for highly nonlinear systems (LIME); socioeconomic barriers and data scarcity.

Table 3 summarises six distinct applications of XAI in Predictive PdM across various industrial sectors. This discussion advances from a descriptive summary to a critical synthesis of the insights obtained, the methodological rigour employed, and the efficacy of various XAI techniques in meeting essential operational demands, thereby achieving greater analytical depth.

The case studies illustrate the versatility of XAI while highlighting challenges related to data quality, model complexity, and the need to tailor explanations to different users.

1. Techniques and Methodological Strengths of XAI: A significant distinction is present between the model-agnostic methods (SHAP, LIME, CFE) prevalent in regulated and numerical domains, and the domain-specific strategies created for particular classification problems:

SHAP serves as the principal tool for delivering both local and global explanations across various domains, including Nuclear Generation Asset, Smart Agriculture, and Gas Turbine Prognostics. LIME also demonstrates versatility in these applications. The prominence of this approach is primarily attributed to its mathematical foundation and model-agnostic flexibility. The Smart Agriculture case utilised LIME and SHAP in conjunction with Counterfactual Explanations (CFE) to offer insights across four specific dimensions of explainability: data, model, outcome, and end-user. Two cases presented innovative, domain-specific methodologies for explainable artificial intelligence (XAI). The Aviation Maintenance study introduced Failure Diagnosis Explainability (FDE), utilising Random Forest, a "white box" algorithm, to evaluate failure characteristics relative to expected distributions via visual box plots and a numerical chi-factor. This ensured transparency by establishing trust grounded in knowledge — a necessary component of critical aerospace diagnostics. The Automotive Applications case employed a novel framework in which a Neural Network produced a Statistical Distance Pseudometric analogous to the Wasserstein distance. This architecture established a clear connection between complex image data — specifically, wear patterns on air filters — and a quantifiable metric used in a Bayesian regression model to estimate wear levels and predict lifetime.

2. Examining Trust, Uncertainty, and Actionability: The capacity of XAI to deliver actionable insights and enhance trust is a key takeaway from these industrial applications, accomplished mainly through the integration of uncertainty quantification and the production of user-centric outputs.

Uncertainty Quantification (UQ) is essential in high-risk environments, where trust is contingent on both the rationale for a decision and the confidence level of the model supporting it. The Gas Turbine case illustrated this rigour through the integration of SHAP and Deep Learning Uncertainty (Aleatoric Uncertainty, AU). The AU functioned as both an anomaly indicator through CUSUM detection and a means to enhance prognostic explanations by indicating the model's confidence. The Automotive case used Bayesian Regression to obtain 95% credible and predictive intervals for wear estimates while effectively accounting for the sensor apparatus's measurement error.

- **Actionable Insights versus Debugging:** XAI should advance from solely debugging models to delivering specific recommendations for operational improvements. The Sustainable Urban Water Systems case presented specific, actionable metrics obtained from interpretable models: a potential 20% reduction in energy consumption through optimised pump scheduling and a 12% decrease in water loss from leak detection initiatives. The Smart Agriculture case employed Counterfactual Explanations (CFEs) to present end-users with specific "what-if" scenarios, outlining the minimal alterations to input features (e.g., specific error counts or flow rates) needed to attain a desired output (e.g., preventing a failure classification).
 - **Accessibility for End Users:** The Nuclear Generation Asset study emphasised the importance of converting technical SHAP values into comprehensible text-based explanations for operational staff lacking machine learning expertise, which is essential for assisting decision-makers in regulated industries with the implementation of prognostic models.
- 3. Context of Data and Empirical Rigour:** The case studies demonstrate ongoing data limitations and present emerging solutions for enhancing methodological robustness.
- **Scarcity of Data and Synthesis:** A standard limitation in Aviation Maintenance (small failure dataset of 144 events), Machines in Production (generic, experimental data), and Gas Turbines (lack of real-world failure data) is the difficulty in acquiring large, labelled, real-world failure datasets. The Automotive case used spatial block bootstrapping to generate synthetic training data for imaging models, addressing the high cost of collecting labelled physical wear data.
 - **Assessment and Verification:** The Gas Turbine case demonstrated empirical rigour by validating the quality of the XAI method based on the established properties of local accuracy and consistency as derived from Shapley values. This verification step is essential for the field, given the insufficient explanation of assessment methods in PHM-XAI identified in the literature.
 - **Performance Benchmarking:** Quantitative results indicated measurable improvements when XAI techniques were integrated into the model pipeline. The XGBoost classifier in the Smart Agriculture application demonstrated enhancements of 7.09% in F1 score and 10.66% in accuracy relative to comparable studies. The Gas Turbine case demonstrated that employing SHAP global explanations for feature selection improved prognostic performance, yielding results comparable to the leading published techniques in the field.

The analysis of these case studies highlights the necessity of context-aware XAI integration. Successful implementation requires selecting XAI methods that align with the specific PdM task and the end user's technical capabilities, as summarised by the following table (Table 4).

Table 4

Actionable learning from the required XAI approach for the PdM task category

PdM Task Category	Primary XAI Approach Required	Actionable Learning
Diagnostics (Classification/Diagnosis) (Aviation, Manufacturing)	Model-Specific/Transparent (e.g., Random Forest, FDE, explicit rule generation)	Focus on pinpointing Root Cause ("Why") and validating prediction against historical norms (chi-factor, boxplots).
Prognostics (RUL/Wear Estimation) (Gas Turbine, Automotive)	Uncertainty Quantification (Bayesian models, AU/EU) and Feature Attribution (SHAP)	Focus on Model Confidence and Action Planning (prediction intervals, confidence trends). Use SHAP global insights for feature engineering to boost core performance.
Operational Optimisation (Urban Water, Smart Agriculture)	Counterfactuals (CFE/DiCE) and Model-Agnostic Attribution (SHAP/LIME)	Focus on Quantifiable Impact (energy/cost savings, resource allocation) and on providing prescriptive scenarios for end users.

6. Proposed Methodological Framework and XPA for XAI in PdM

The integration of XAI within PdM is critical for bridging the gap between sophisticated black-box models and human decision-making, particularly concerning trustworthiness and actionable insights. Drawing on the literature, we can define the standard performance metrics for XAI assessment and outline a generic methodological approach for implementing XAI in PdM.

6.1. Key Performance Metrics for Assessing XAI in PdM Applications

Assessing XAI involves a blend of technical performance metrics (model fidelity and speed) and human-centric metrics (trust, interpretability, and alignment with domain knowledge).

a. Generic XAI Evaluation Framework (XPA)

A comprehensive set of Explainability Parameters (XPA) and their measurement criteria is proposed to standardise XAI assessment (see Table 5) as illustrated by Fig. 9:

Table 5

Generic XAI Evaluation Framework

Metric Category	Qualitative Metrics	Quantitative Metrics
Explanation Scope	Depth of Explanation: Local (specific region), Global (entire conditional distribution), or Both.	Model Coverage: Percentage of instances covered by the explanation (All or Individual).
Model Fidelity & Stability	Approximation: Ability to closely explain the black-box model output (High, Medium, Low). Stability: Consistent outcome when comparing explanations between similar instances within the same model (High, Medium, Low).	Consistency: Ability to explain consistently between different models (High, Medium, Low). Local Accuracy: Confirmation that the sum of feature contributions equals the prediction minus the average prediction.
Feature Contribution	Abnormality detection (High, Medium, Low).	Feature Importance: Measured as a percentage contribution to the decision.
Trust & Compliance	Bias in prediction (High, Medium, Low).	Privacy (K): Compliance check (Yes, No).
Structural Clarity	Decomposability: Ability to explain the model, including input, output, and prediction (High, Medium, Low).	N/A



Fig. 9. XAI Evaluation Metrics

a. Metrics Specific to PdM Application Types

The importance and type of metrics shift depending on whether the PdM task is focused on immediate diagnosis or long-term prognosis. Explainable models often generate metrics that directly address the uncertainty, cost, or time implications of predictions, as described in Table 6 and illustrated in Fig. 10.

Table 6

Metrics Specific to PdM Application Types

PdM Type	Application	Common Quantitative Metrics	Specific XAI Metrics/Techniques
Prognostics (RUL/TTF)		RMSE (Root Mean Squared Error): Measures the spread between the prediction and actual values.	95% Credible/Prediction Intervals: Bayesian regression models use these to estimate wear levels and the reliability of predictions.
		Early Prognostic Metric (S): Gives higher scores for errors in early prediction compared to late prediction, reflecting prognostic quality.	Confidence Indicators: Deep learning Aleatoric Uncertainty (AU) measures can reflect model confidence, often decreasing as RUL estimation nears failure.
Diagnostics (Fault ID, Classification)		Accuracy/F1 Score/ROC-AUC: Standard metrics evaluating classification correctness, especially important against a naive "most frequent" baseline. Consistency: Ensuring that explanations remain stable when similar inputs are provided to the model.	Chi-factor: Measures the extent to which a new failure's feature values match the expected distribution (boxplots) of a specific diagnosis (class). CUSUM (Cumulative Sum) Detection: Used in conjunction with deep learning uncertainty to flag abrupt anomalous observations.

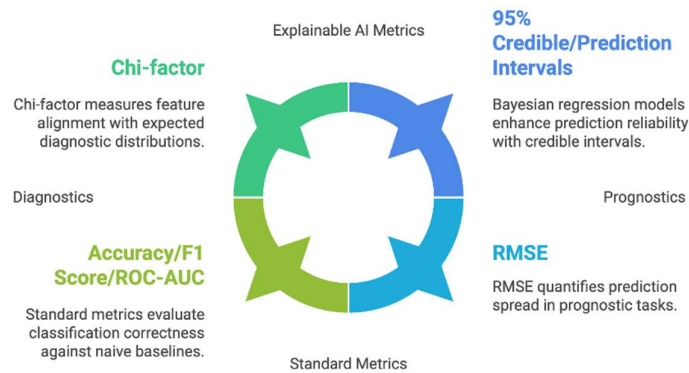


Fig. 10. PdM Metrics and Techniques

6.2. Generic Approach for Using XAI in PdM

A generic approach for integrating XAI into PdM typically follows a structured pipeline involving data preparation, model training and selection, and a specialised diagnosis/explanation phase. This roadmap uses a phased breakdown Input-Activities-Proposed Tools/Techniques-Output) synthesising insights across different industrial contexts.

The process of XAI involves three phases (see Fig. 11): data pre-processing and feature engineering, model training and selection, and explanation generation and action validation.

The first phase involves data cleansing, labelling, feature engineering, and historical reconstruction to align maintenance and usage data. The second phase focuses on selecting models suitable for explainability, typically yielding supervised multi-classification problems. The third phase involves generating raw explanation data and translating it into actionable insights using XAI methods such as SHAP and LIME, aiming to link predictions to required actions.

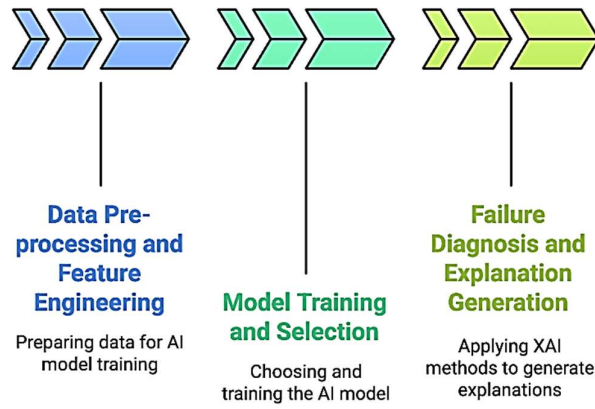


Fig. 11. XAI integration in the PdM approach

- **Phase 1: Data Pre-processing and Feature Engineering**

This foundational phase focuses on integrating heterogeneous data sources and preparing them for the AI model, including generating features representative of component usage (see Table 7).

Table 7

Phase 1 of the proposed approach

Input	Activities	Proposed Tools/Techniques	Output
Maintenance Data (Textual/Categorical failure reports, repair cards, maintenance history).	Cleaning and Validation: Handling measurement faults, incompleteness, and ensuring data quality.	Data cleaning tools, statistical methods (e.g., Last Observation Carried Forward (LOCF) for time series).	Cleaned and validated datasets.
Usage/Sensor Data (Continuous/Numeric sensor readings, e.g., temperature, vibration, speed).	Labelling: Grouping similar failure descriptions into diagnostic categories (classes).	Domain expertise, automated text mining (future direction).	Labelled Failure Data.
	Feature Engineering: Calculating single values/characteristics (features) from continuous data (e.g., average, standard deviation, kurtosis, RUL/TTF).	Time series feature extraction (lag features, window features, periodic cyclic features).	Feature set reflecting usage and degradation.
	Historical Reconstruction: Aligning maintenance event timelines with usage data across components and assets.	Data fusion techniques, visualisation tools (for preliminary verification).	Combined, the Feature-Rich Dataset is ready for training.

- **Phase 2: Model Training and Selection**

This phase selects and trains the core AI model, prioritising those compatible with subsequent explainability needs (e.g., “white box” or high-fidelity black box models), as illustrated in Table 8.

Table 8

Phase 2 of the proposed approach

Input	Activities	Proposed Tools/Techniques	Output
Combined, Feature-Rich Dataset (labelled training data).	Algorithm Selection: Choosing the most suitable model based on problem type (supervised multi-classification or regression) and preliminary performance.	Supervised ML (Random Forest, XGBoost, SVM, Naive Bayes), Deep Learning (LSTM, CNN, Neural Networks).	Trained AI Model (e.g., Random Forest chosen for highest accuracy in the F-16 case study).
	Model Evaluation: Assessing initial performance against benchmarks to select the final model.	Accuracy, F1 Score, ROC-AUC, comparison against the "most frequent" baseline.	Performance metrics and selection criteria for model transparency.

• Phase 3: Failure Diagnosis and Explanation Generation

This final phase applies XAI methods to the model's output, translating predictions into human-understandable terms for domain experts and end users, as illustrated by Table 9.

Table 9

Phase 3 of the proposed approach

Input	Activities	Proposed Tools/Techniques	Output
Trained AI Model, New Failure/Test Instance Data.	Diagnosis Assessment: Generating a predicted outcome (e.g., fault class, RUL value, anomaly score).	Model inference layer.	Probable failure diagnosis (e.g., bar graph visualisation).
	Explainability Mapping (Fidelity): Applying XAI to the prediction to determine contributing factors.	Model-Agnostic: SHAP, LIME, Counterfactual Explanations (CFE/DiCE).	Raw explanation data (e.g., SHAP values, feature attribution vectors).
	XAI Feature Assessment: Determining variable importance relative to specific diagnoses/classes.	Model-Specific: FDE (Failure Diagnosis Explainability), Variable Importance plots (for "white box" models like Random Forest/Decision Trees).	Top contributing features per class.
	Human-Centric Explanation (End-User): Translating technical explanation outputs into actionable, simple formats.	Visual/Numerical Techniques: Boxplots (comparing new failure features vs. expected distribution), chi-factor (numerical match index), text generation algorithms.	Human-interpretable explanation (Visual plots, numerical scores, actionable counterfactual scenarios).
	Action Validation: Integrating explanation output into decision systems (e.g., prioritising maintenance, scheduling complex repairs).	Domain expertise review, simulation via Digital Twins (future context).	Maintenance recommendation and justification.

7. Conclusion

The review presented in this study highlights the critical role of Explainable Artificial Intelligence in enhancing Predictive Maintenance systems. The findings reveal that XAI significantly improves understanding and decision-making by increasing transparency, trust, and clarity in AI models. It allows stakeholders to comprehend model predictions, identify biases, and foster effective human-AI collaboration. The study introduces a holistic framework, the Explainability Parameters (XPA) Framework, that provides rigorous metrics tailored to various PdM applications. Additionally, a structured three-phase methodological approach is proposed to facilitate the translation of technical outputs into actionable maintenance recommendations.

The actionable implications of these findings emphasise the need for industrial managers and AI practitioners to adopt the XPA framework for the standardised evaluation of XAI solutions. This shift should prioritise actionable insights over mere debugging of model predictions, ensuring that explanations lead to optimal repair plans and stakeholder coordination. Furthermore, the selection of XAI techniques and the presentation of explanations must be context-aware, tailored to specific PdM tasks and the technical expertise of end-users.

Despite these advancements, the review identifies several limitations within the current XAI research landscape in PdM. A significant evaluation deficit exists due to the lack of globally recognised standards for assessing XAI effectiveness, leading to misinterpretations. Additionally, research often focuses narrowly on the explainability of outcomes rather than on practical insights that enhance processes. Limitations related to data quality and generalizability are also noted, as many studies rely on generic or synthetic datasets, hindering the establishment of causal relationships. Furthermore, existing XAI methodologies often inadequately address uncertainties in data and models, thereby compromising the reliability of the explanations they provide.

Looking ahead, future research should aim to address these limitations by developing hybrid, flexible XAI frameworks that integrate multiple techniques to provide more comprehensive explanations. Emphasis should be placed on creating human-centric designs that prioritise knowledge generation and improve communication between humans and AI. Additionally, refining explanatory metrics tied to quantifiable operational outcomes is essential for ensuring practical relevance in PdM applications. Lastly, addressing data integrity challenges will be crucial, with a focus on methodologies that effectively utilise real-world industrial data to overcome the reliance on synthetic datasets.

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